

Bias and Fairness

Bias/Harm sources in ML/AI

Datasheets for datasets

Google Data card playbook/Model Card

By Dr. Ali Algarni

Outlines

- Anonymity
- Security
- Privacy
- Surveillance
- Privacy Protection and Civil Liberties
- Ethical and Legal Framework for Information

Recap

- Ethical Theories
 - Govern and shape the ethics, law, and code of conduct, code of eth
- Code of Ethics vs Code of Conduct
 - govern decision making vs govern actions
- Decision Making Framework
 - Guidelines and code of conducts (or ethics)
- Society, Morality, Ethics and Law
 - Digital and cyber societies/communities vs traditional

Bias and Fairness

Joy Buolamwini & Timnit Gebru: gendershades.org

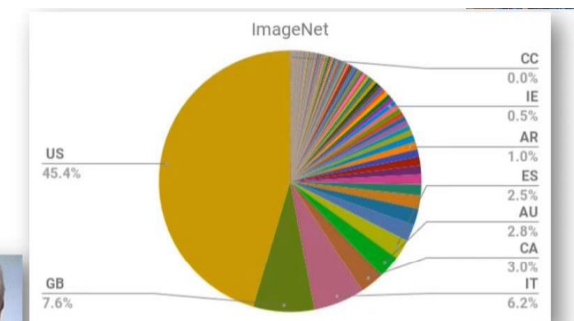
Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE**	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%



Joy Buolamwini & Timnit Gebru, gendershades.org



4.4% of IJB-A images are dark-skinned women



2/3 of ImageNet images from the West (Shankar et al, 2017)

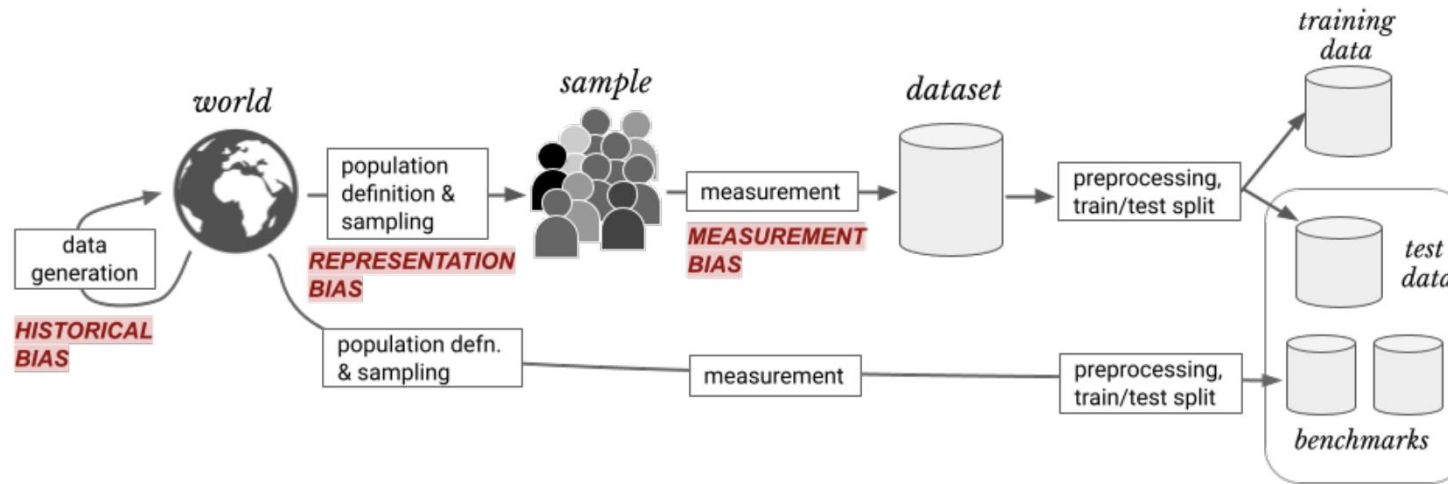
Evaluation Bias

A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle

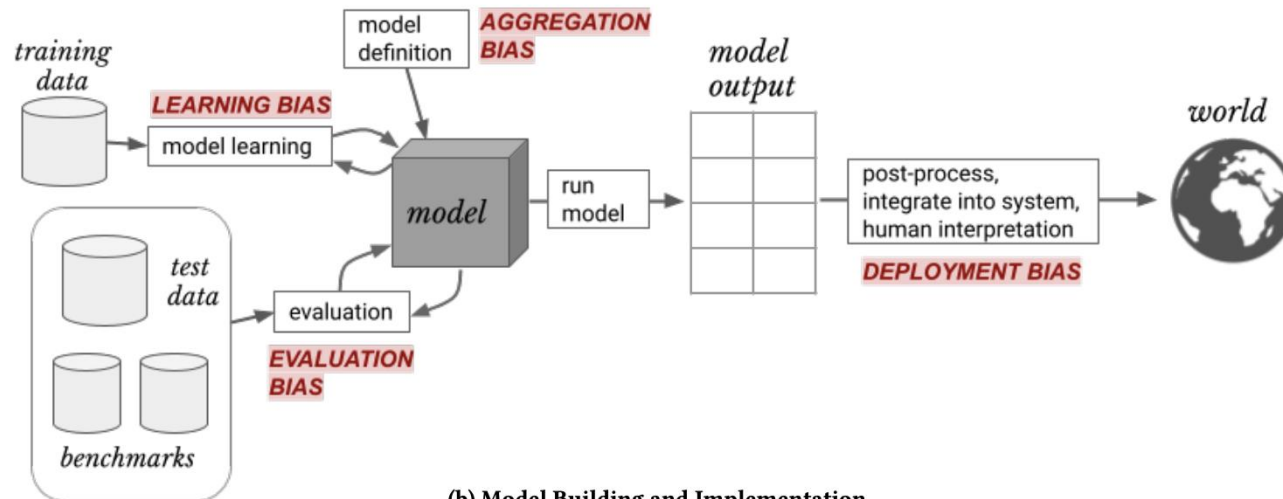
Seven Sources of Harm In Machine learning:

- **Data generation**
 - **Historical Bias** (Word Embedding).
 - **Representation Bias** (Population Sampling, Geographic Diversity in Image Dataset).
 - **Measurement Bias** (Feature Selection, Risk Assessments in the Criminal Justice System).
- **Model Building**
 - **Aggregation Bias** (One-size-fits-all model, Social Media Analysis).
 - **Learning Bias** (Accuracy, pruning).
 - **Evaluation Bias** (Benchmark data for use population, Commercial Facial Analysis Tools).
 - **Deployment Bias** (Risk Assessment Tools in Practice)

A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle



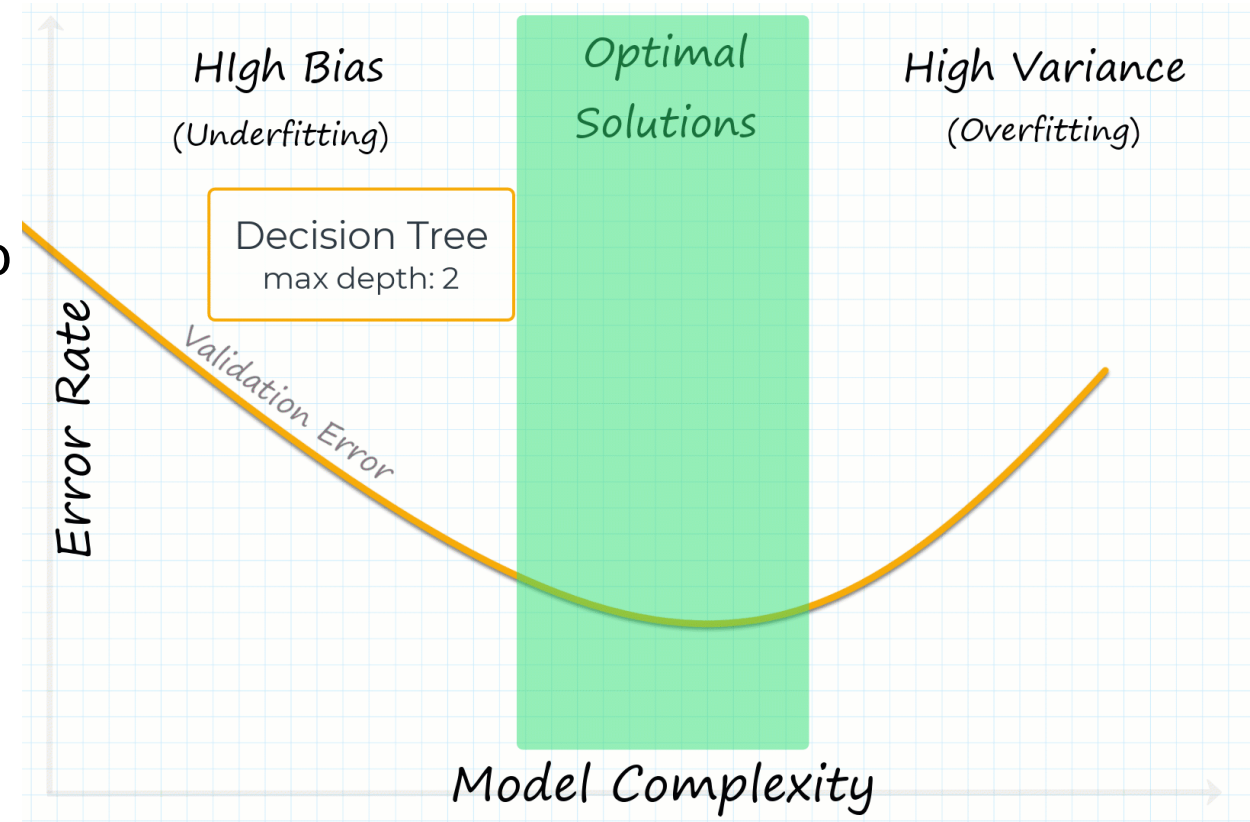
(a) Data Generation



(b) Model Building and Implementation

Example: ML/AI Model Bias and variance

- Two core sources of prediction error in ML
- **Bias** is error from oversimplifying, causing underfitting (missing patterns)
- **Variance** is error from extreme sensitivity to training data, causing overfitting (learning noise).
- Balancing them is essential to minimize total error.
- Ensuring that algorithms do not unfairly harm, discriminate against, or produce untrustworthy results for individuals or groups



Datasheets for datasets.

- **Paper published in 2021**

- *Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Iii, H. D., & Crawford, K. (2021). Datasheets for datasets. Communications of the ACM, 64(12), 86-92.*

- Standardize process for documenting datasets and solve severe consequences when machine learning models are used in high-stakes domains.
- Address the needs of two key stakeholder groups: dataset creators and dataset consumers
- **Transparency of dataset creators** is necessary for **dataset consumers** to be select appropriate datasets sufficiently
 - avoid unintentional misuse

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Datasheets for datasets.

- Every dataset be accompanied with a datasheet that documents
 - Motivation
 - Composition
 - collection process
 - recommended uses
 - Distribution
 - maintenance and so on.
- Datasheets for datasets have:
 - The potential to increase transparency and accountability within the machine learning community,
 - Mitigate unwanted biases in machine learning systems,
 - Facilitate greater reproducibility of machine learning results, and
 - Help researchers and practitioners select more appropriate datasets for their chosen tasks.

Google: The Data Cards Playbook

- Developed by **Google Research**
- Key Objectives
 - **Transparency:** Providing a clear "nutrition label" for datasets.
 - **Responsible AI:** Identifying biases, limitations, and ethical considerations before a model is ever trained.
 - **Usability:** Making complex data information accessible to non-technical stakeholders (policy experts, legal teams, and researchers)
- The Framework:
 - **Ask:** questions different stakeholders need answered.
 - **Inspect:** Evaluating existing data to see if it meets those needs.
 - **Answer:** Filling out the Data Card template with context and technical details.
 - **Audit:** Reviewing the completed card for accuracy, clarity, and completeness.

Google: The Data Cards Playbook



Pushkarna, M., Zaldivar, A., & Kjartansson, O. (2022, June). Data cards: Purposeful and transparent dataset documentation for responsible ai. In *Proceedings of the 2022 ACM conference on fairness, accountability, and transparency* (pp. 1776-1826).

Google Model Card

- Simple, structured overviews of how an advanced AI model was designed and evaluated,
- Supporting Google's [approach to responsible AI](#).
- Each card include ethical issues related to use the model
 - Model Training and Development Evaluation
 - Human Red Teaming
 - Risks and Mitigations
 - Safety Policies
 - Ethics & Safety Reviews

Google Model Card

- Example: **Gemini 3 Pro Model Card**

Ethics and Content Safety

Evaluation Approach: Gemini 3 Pro was developed in partnership with internal safety, security, and responsibility teams. A range of evaluations and red teaming activities were conducted to help improve the model and inform decision-making. These evaluations and activities align with [Google's AI Principles](#) and [responsible AI approach](#), as well as Google's Generative AI policies (e.g. [Gen AI Prohibited Use Policy](#) and the [Gemini API Additional Terms of Service](#)).

Evaluation types included but were not limited to:

- **Training/Development Evaluations** including automated and human evaluations carried out continuously throughout and after the model's training, to monitor its progress and performance;
- **Human Red Teaming** conducted by specialist teams who sit outside of the model development team, across the policies and desiderata, deliberately trying to spot weaknesses and ensure the model adheres to safety policies and desired outcomes;
- **Automated Red Teaming** to dynamically evaluate Gemini for safety and security considerations at scale, complementing human red teaming and static evaluations;
- **Ethics & Safety Reviews** were conducted ahead of the model's release

In addition, we perform testing following the guidelines in [Google DeepMind's Frontier Safety Framework](#) (FSF).

Source: Gemini 3 Pro Model Card

<https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-Pro-Model-Card.pdf>

More resources:

- Data Card Playbook Google:
<https://sites.research.google/datacardsplaybook/>
- Data Cards Playbook Github community:
 - <http://github.com/PAIR-code/datacardsplaybook>
- Google Model Cards:
 - <https://modelcards.withgoogle.com/model-cards>